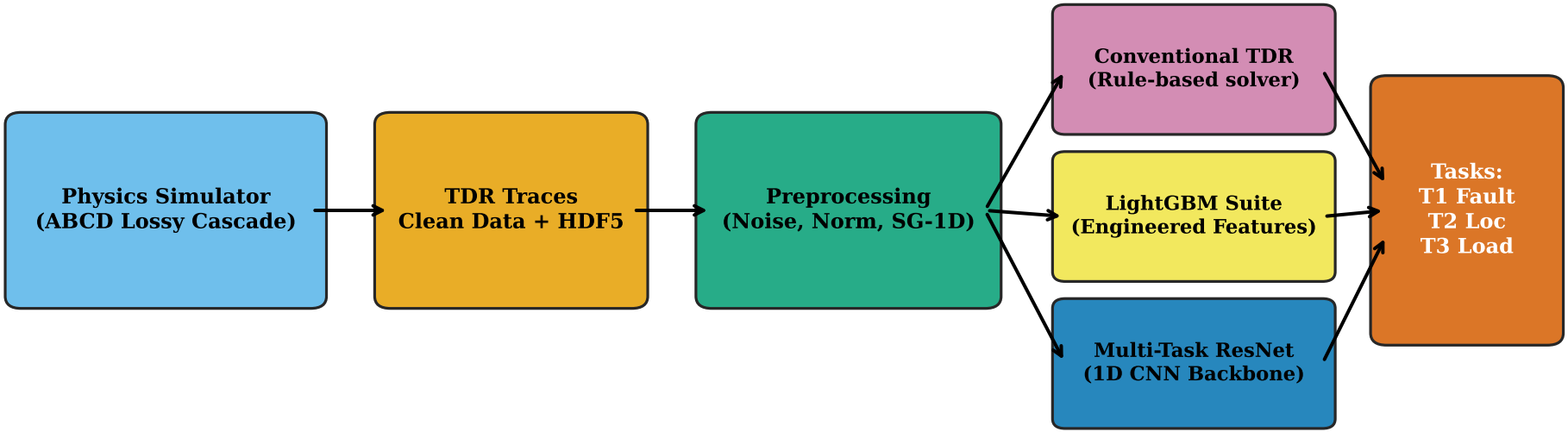
Physics Simulator
(ABCD Lossy Cascade)
TDR Traces
Clean Data + HDF5
Preprocessing
(Noise, Norm, SG-1D)
Conventional TDR
(Rule-based solver)
LightGBM Suite
(Engineered Features)
Multi-Task ResNet
(1D CNN Backbone)
Tasks:
T1 Fault
T2 Loc
T3 Load